DATA STRUCTURES (C) & OOP (C++) PROJECT PROPOSAL

PROJECT AND TEAM INFORMATION

Project Title

Library Management System

Student / Team Information

Team Name: Library Management Team	
Team Member 1 (Team Lead) Name, Student ID, Roll No, Email	Varun Panwar Student ID: 2510012520 Roll No: 2028262 Email: 2510012520@geu.ac.in 
Team Member 2 Name, Student ID, Roll No, Email	Shubh Saini Student ID: 2510011685 Roll No: 2028159 Email: 2510011685@geu.ac.in 
Team Member 3 Name, Student ID, Roll No, Email	Nilesh Yadav Student ID: 2510370404 Roll No: 2027945 Email: 2510370404@geu.ac.in 

PROPOSAL DESCRIPTION

Motivation

Managing a library becomes harder when the number of books, students, and daily transactions keeps increasing. If everything is recorded in separate registers, it can take a lot of time to find information and mistakes can happen easily. Our Library Management System is planned as a simple application that keeps the main library activities in one place.

The system will help with adding and managing books and members, searching for books, keeping issue and return records, and checking due dates and availability. We will use suitable Data Structures such as arrays, linked lists, queues, stacks, trees, and hashing wherever they make sense. C++ classes and OOP concepts will be used to keep the program organized and easier to manage.

Through this project, we also want to get practical experience in using Data Structures and Object-Oriented Programming to solve a problem that is commonly seen in a library.

State of the Art / Current Solution

Many libraries already use computer-based systems to keep track of books and members. These systems can usually search the book catalogue, record issue and return activities, and show whether a book is available. However, smaller libraries may still use registers or simple spreadsheets for these tasks.

Manual records and basic spreadsheets can work for a small collection, but searching and updating information can become slower as the number of books and transactions increases. Our proposed system will bring the main library operations together in one application. The main focus of this project is to understand how C++ Data Structures and OOP can be used in a practical library system.

PROJECT GOALS AND MILESTONES

Project Goals

1. Build a C++ based Library Management System to keep book and member records in an organized way.
2. Add features for adding, updating, deleting, and displaying book details.
3. Make it easy to search for books by using suitable Data Structures.
4. Record book issue and return activities and keep the availability status updated.
5. Use OOP concepts such as classes, objects, encapsulation, constructors, and member functions.
6. Use Data Structures such as linked lists, queues, stacks, trees, or hashing where they are useful for the system.

Project Milestones

1. Finalize the requirements, main features, and basic design of the system.
2. Plan the book, member, and transaction data models in C++.
3. Develop the basic book and member management features.
4. Add searching and sorting using suitable Data Structures.
5. Add book issue, return, due-date, and availability features.
6. Combine all modules, test the complete program, fix errors, and make the system easier to use.

Project Approach

We will build the Library Management System step by step in C++. First, we will identify the main parts of the system and divide the work into modules such as Book Management, Member Management, Search, Issue and Return, and Report Management.

Classes such as Book, Member, Transaction, Library, and LibraryManager will represent the main parts of the program. OOP concepts will help us keep the code structured and make individual parts easier to update.

For storing and handling records, we will choose Data Structures according to the requirement of each operation. For example, linked lists can be useful for records that change in size, queues can be used for waiting requests, stacks can store selected history information, trees can help with ordered searching, and hashing can provide quick lookups.

We will first develop and test each module separately. After that, the modules will be connected into one application. Testing will mainly check record handling, searching, issue and return operations, and the overall working of the system.

Project Phases

Phase 1: Planning & Requirement Analysis

Understand the problem, decide the objectives and scope, list the requirements, and select the C++ and Data Structures needed for the project.

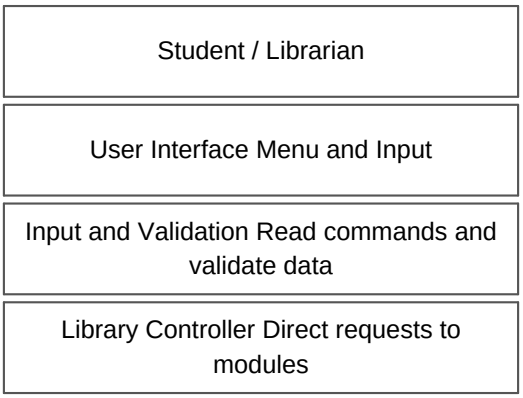
Phase 2: System Design & Development

Design the main modules and data handling structure, then develop the library operations using C++ and the selected Data Structures.

Phase 3: Testing & Final Implementation

Test different book, member, and transaction cases, correct errors, check that all operations work properly, prepare reports, and complete the final documentation and demonstration.

SYSTEM ARCHITECTURE (HIGH LEVEL DIAGRAM)



Book Management	Member Management
Add, update, delete, display	Register and manage members

Transaction Management	Search and Sort
Issue, return and due tracking	Find books and members

Data Structures	File Storage
Linked List, Queue, Stack, Tree, Hashing	Save books, members and transactions

Reports and Output	System Status
Availability, issued books, member and transaction reports	Success messages and error handling

Project Outcome / Deliverables

The main outcome of the project will be a C++ Library Management System that brings book management, member management, searching, issue and return processing, availability tracking, and basic reporting together in one application.

The system will use a simple menu-based interface so that a librarian or authorized user can carry out common library tasks. Records will be kept in an organized form, while the project will also demonstrate how Data Structures and OOP concepts can be used together.

Assumptions

1. The system is designed mainly for a small or medium-sized library.

2. Users are expected to enter basic details correctly, such as book ID, title, author, member ID, and transaction information.

3. For this project, library records will be stored locally during the demonstration.

4. The first version will focus on book management, member management, search, issue, return, and basic reports.

REFERENCES

1. M. T. Goodrich, R. Tamassia, and M. H. Goldwasser, Data Structures and Algorithms in C++.

2. Robert Lafore, Object-Oriented Programming in C++.

3. Bjarne Stroustrup, Programming: Principles and Practice Using C++.

4. Seymour Lipschutz, Data Structures, Schaum's Outline Series.

PROJECT SUMMARY

The Library Management System is a C++ project based on Data Structures and OOP. It is designed to make common library tasks easier by keeping book and member information together, supporting book searches, recording issue and return activities, and showing useful status information.

The project is also intended to give us practical experience with C++ classes, OOP concepts, and Data Structures. By combining these concepts in one application, we can understand how they can be used to solve a real-world problem in a simple and organized way.